Preliminary Visualizations: U.S. Infrastructure, Jobs and Poverty

Objective

This project explores whether infrastructure construction spending and the number of jobs can help predict poverty rates in the United States. We use annual data from 2002 to 2024 to examine the relationships between these variables.

Load Data

First, we load the required packages and the cleaned dataset from the previous step.

```
source("00_requirements.R")
source("02_Functions.R")

data <- read.csv("data/processed/us_annual_2002_2024.csv")

dir.create("figures", showWarnings = FALSE)
dir.create("results", showWarnings = FALSE)

head(data)
```

```
##   year construction_nominal_musd jobs_thousands cpi_index poverty_pct
## 1 2002                     57483       130633.8  179.8667        12.1
## 2 2003                     57139       130325.2  184.0000        12.5
## 3 2004                     58624       131755.3  188.9083        12.7
## 4 2005                     64140       134023.1  195.2667        13.3
## 5 2006                     72039       136432.8  201.5583        13.3
## 6 2007                     76682       137978.4  207.3442        13.0
##   construction_real_2024_musd jobs_millions
## 1                   100253.77      130.6337
## 2                    97415.21      130.3252
## 3                    97350.08      131.7553
## 4                   103041.65      134.0231
## 5                   112118.92      136.4327
## 6                   116014.85      137.9784
```

1. Summary Statistics

We first examine the basic information about poverty rates, construction spending, and employment.

```
summary(data)
```

```
##       year      construction_nominal_musd jobs_thousands     cpi_index
##  Min.   :2002   Min.   : 57139            Min.   :130325   Min.   :179.9
```

```
##  1st Qu.:2008   1st Qu.: 78003            1st Qu.:132970   1st Qu.:211.0
##  Median :2013   Median : 82529            Median :137978   Median :233.0
##  Mean   :2013   Mean   : 88402            Mean   :140369   Mean   :234.7
##  3rd Qu.:2018   3rd Qu.: 96296            3rd Qu.:146447   3rd Qu.:253.4
##  Max.   :2024   Max.   :148337            Max.   :157694   Max.   :313.7
##   poverty_pct    construction_real_2024_musd jobs_millions
##  Min.   :11.90   Min.   : 97350              Min.   :130.3
##  1st Qu.:12.55   1st Qu.:110344              1st Qu.:133.0
##  Median :13.20   Median :116015              Median :138.0
##  Mean   :13.57   Mean   :116544              Mean   :140.4
##  3rd Qu.:14.50   3rd Qu.:121589              3rd Qu.:146.4
##  Max.   :15.90   Max.   :148337              Max.   :157.7
```

```r
variables <- data[, c(
  "poverty_pct",
  "construction_real_2024_musd",
  "jobs_millions"
)]

summary(variables)
```

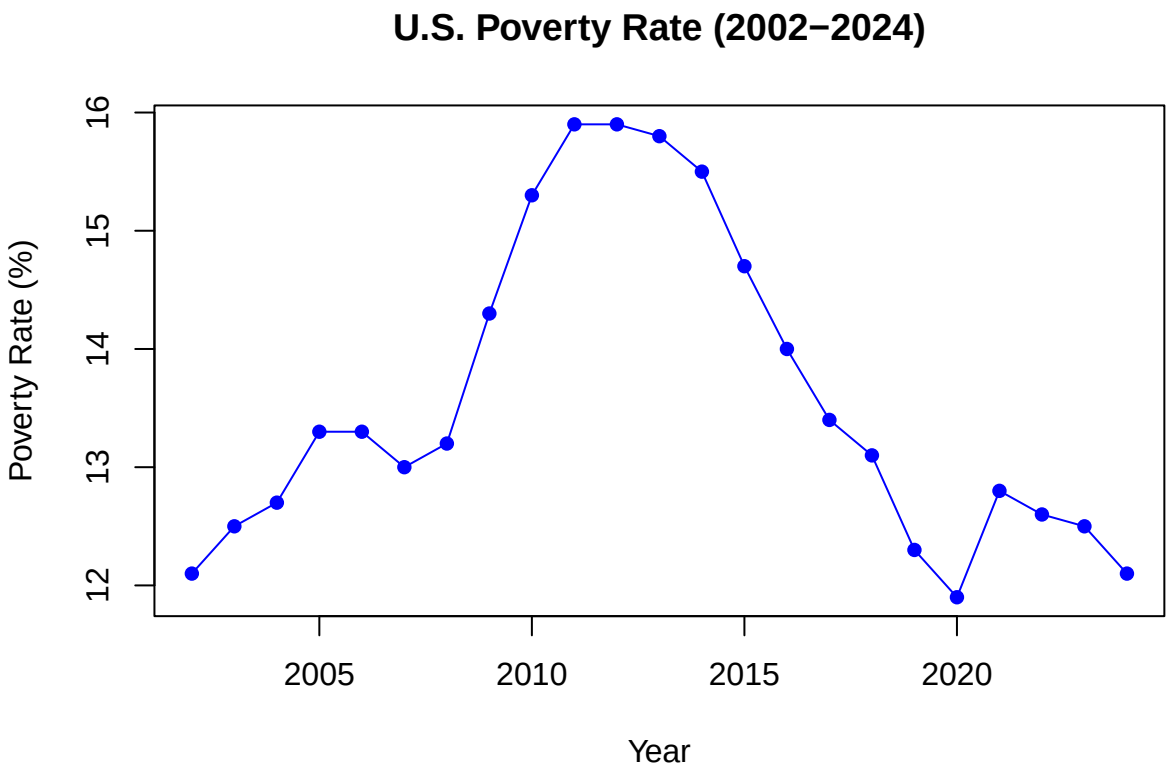
```
##   poverty_pct    construction_real_2024_musd jobs_millions
##  Min.   :11.90   Min.   : 97350              Min.   :130.3
##  1st Qu.:12.55   1st Qu.:110344              1st Qu.:133.0
##  Median :13.20   Median :116015              Median :138.0
##  Mean   :13.57   Mean   :116544              Mean   :140.4
##  3rd Qu.:14.50   3rd Qu.:121589              3rd Qu.:146.4
##  Max.   :15.90   Max.   :148337              Max.   :157.7
```

The dataset contains 23 annual observations from 2002 to 2024. The poverty rate ranged from 11.9% to 15.9%, with an average of 13.57%. Real construction spending ranged from approximately $97.35 billion to $148.34 billion, while employment ranged from 130.3 million to 157.7 million jobs. These statistics show that all three variables changed over the study period, which allows us to examine their relationships.

2. Poverty Rate Over Time

We use a line graph to show how the poverty rate changed in the United States between 2002 and 2024.

```r
plot(
  data$year,
  data$poverty_pct,
  type = "o",
  pch = 16,
  col = "blue",
  main = "U.S. Poverty Rate (2002-2024)",
  xlab = "Year",
  ylab = "Poverty Rate (%)"
)
```



The poverty rate increased from 12.1% in 2002 to a peak of 15.9% in 2011. After 2011, the poverty rate generally declined, reaching its lowest point of 11.9% in 2020. There was a small increase in 2021, but the rate remained relatively low compared with the early 2010s. In 2024, the poverty rate was 12.1%, the same as in 2002. This shows that poverty did not follow a constant upward or downward trend during the study period.

3. Construction Spending and Poverty

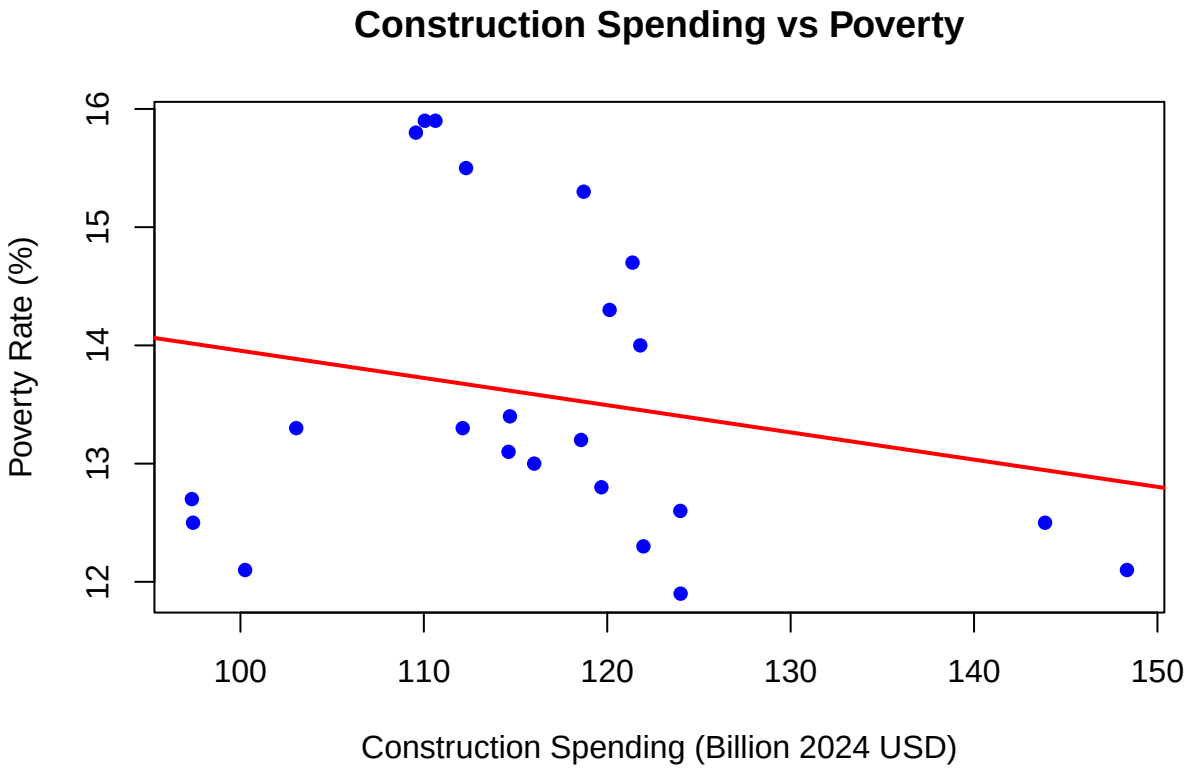
We create a scatterplot to examine the relationship between infrastructure construction spending and poverty rates.

```
construction <- data$construction_real_2024_musd / 1000

plot(
  construction,
  data$poverty_pct,
  pch = 16,
  col = "blue",
  main = "Construction Spending vs Poverty",
  xlab = "Construction Spending (Billion 2024 USD)",
  ylab = "Poverty Rate (%)"
)

model1 <- lm(data$poverty_pct ~ construction)

abline(model1, col = "red", lwd = 2)
```



The scatterplot shows a weak negative relationship between real construction spending and poverty rates. The correlation coefficient is -0.214, which suggests that higher construction spending is generally associated with slightly lower poverty rates. However, the points are widely scattered around the regression line. This means construction spending alone does not have a strong linear relationship with poverty. The graph suggests that other economic factors may play an important role in explaining poverty rates.

4. Employment and Poverty

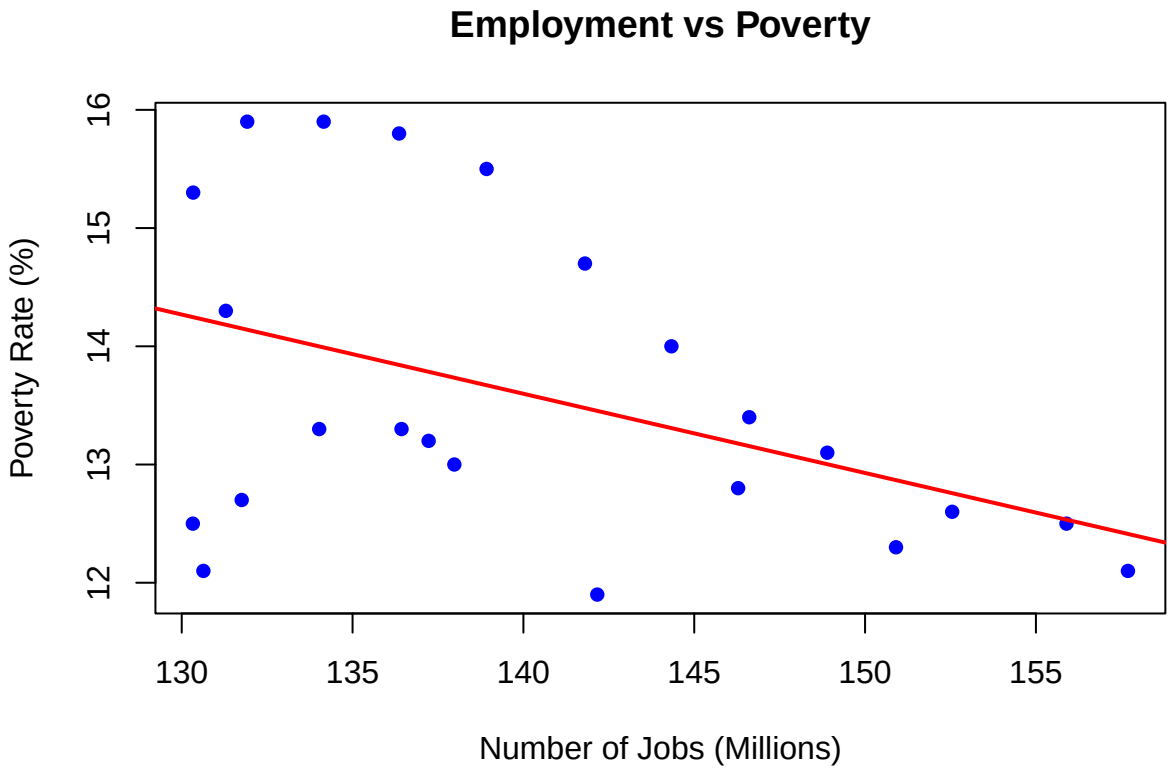
We create another scatterplot to examine the relationship between the number of jobs and poverty rates.

```
plot(
  data$jobs_millions,
  data$poverty_pct,
  pch = 16,
  col = "blue",
  main = "Employment vs Poverty",
  xlab = "Number of Jobs (Millions)",
  ylab = "Poverty Rate (%)"
)

model2 <- lm(
  poverty_pct ~ jobs_millions,
  data = data
```

```
)

abline(model2, col = "red", lwd = 2)
```

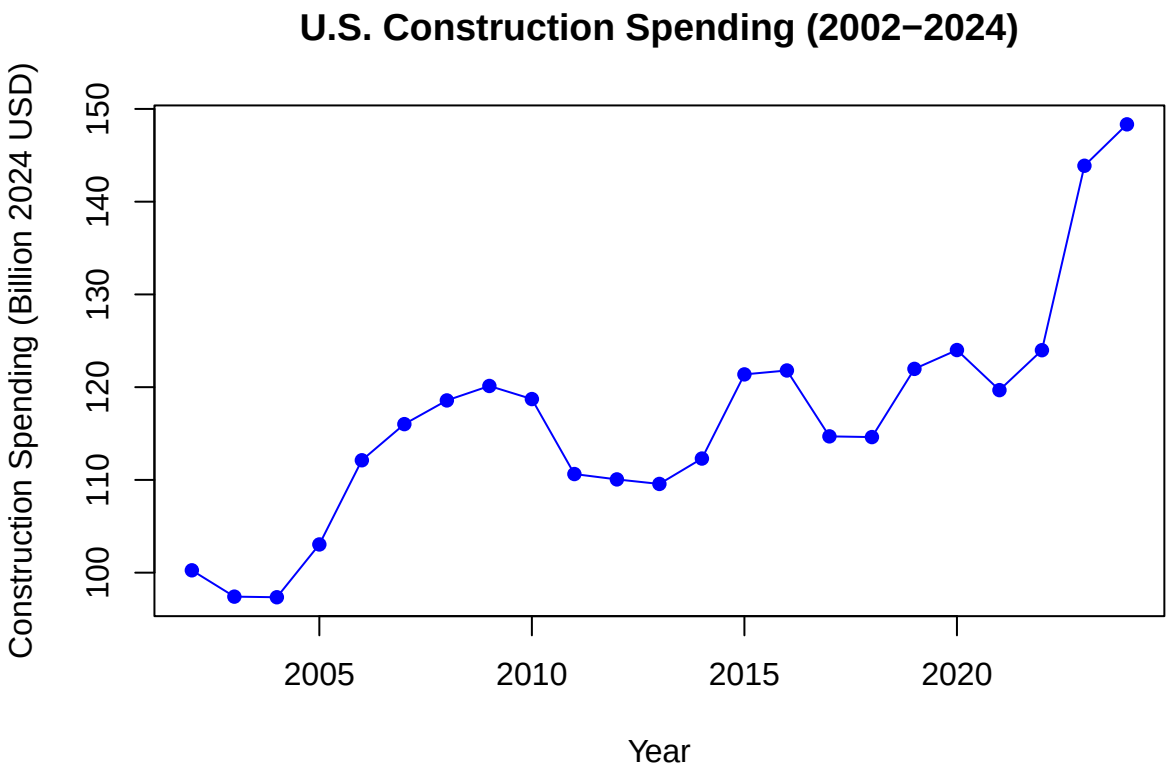


The scatterplot shows a moderate negative relationship between employment and poverty rates. The correlation coefficient is -0.434, indicating that years with more jobs generally had lower poverty rates. The downward regression line supports this relationship. Compared with construction spending, employment has a stronger negative correlation with poverty. However, the points still show considerable variation, suggesting that the number of jobs alone cannot fully explain changes in poverty rates.

5. Construction Spending Over Time

We use a line graph to examine how construction spending changed between 2002 and 2024.

```
plot(
  data$year,
  construction,
  type = "o",
  pch = 16,
  col = "blue",
  main = "U.S. Construction Spending (2002-2024)",
  xlab = "Year",
  ylab = "Construction Spending (Billion 2024 USD)"
)
```

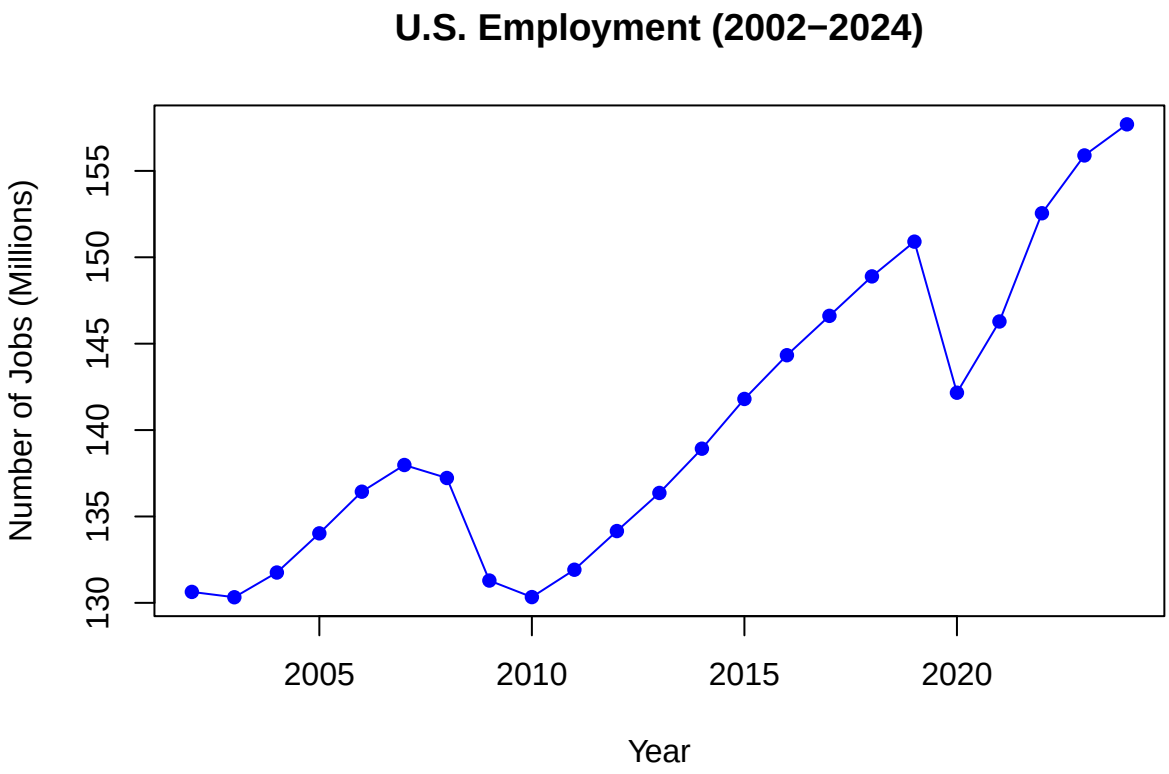


Real construction spending increased from approximately $100.25 billion in 2002 to $148.34 billion in 2024. Spending grew during the mid-2000s, declined after 2009, and remained relatively stable during much of the 2010s. After 2021, construction spending increased sharply and reached its highest level in 2024. Since these values are adjusted for inflation, the graph shows changes in real spending rather than changes caused only by rising prices.

6. Employment Over Time

This graph shows changes in the number of payroll jobs in the United States.

```
plot(
  data$year,
  data$jobs_millions,
  type = "o",
  pch = 16,
  col = "blue",
  main = "U.S. Employment (2002-2024)",
  xlab = "Year",
  ylab = "Number of Jobs (Millions)"
)
```



Employment generally increased from approximately 130.6 million jobs in 2002 to 157.7 million in 2024. However, the graph shows two noticeable declines. The first occurred around 2008–2010, during the Great Recession. The second occurred in 2020, during the COVID-19 pandemic. Employment recovered after both declines and reached its highest level in 2024. These changes suggest that employment is sensitive to major economic disruptions.

7. Correlation Analysis

We calculate correlations to examine the strength and direction of the linear relationships between our variables.

```
cor_matrix <- cor(variables)

round(cor_matrix, 3)
```

```
##                             poverty_pct construction_real_2024_musd
## poverty_pct                       1.000                      -0.214
## construction_real_2024_musd      -0.214                       1.000
## jobs_millions                    -0.434                       0.791
##                             jobs_millions
## poverty_pct                        -0.434
## construction_real_2024_musd         0.791
## jobs_millions                       1.000
```

```r
write.csv(
  cor_matrix,
  "results/correlation_matrix.csv"
)
```

The correlation matrix shows three important relationships. First, construction spending and poverty have a weak negative correlation of -0.214. Second, employment and poverty have a moderate negative correlation of -0.434. This suggests that employment has a stronger linear relationship with poverty than construction spending does. Third, construction spending and employment have a strong positive correlation of 0.791. This indicates that the two economic variables often increase together. However, these correlations do not establish causation, and common trends over time may influence the results.

8. Preliminary Regression Analysis

We use a multiple linear regression model to examine how construction spending and employment are jointly associated with poverty rates.

```r
model <- lm(
  poverty_pct ~ construction_real_2024_musd + jobs_millions,
  data = data
)

summary(model)
```

```
##
## Call:
## lm(formula = poverty_pct ~ construction_real_2024_musd + jobs_millions,
##     data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9309 -0.7789 -0.3248  0.6201  2.0480
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  2.457e+01  4.504e+00   5.455 2.44e-05 ***
## construction_real_2024_musd  3.724e-05  3.450e-05   1.079   0.2933
## jobs_millions               -1.093e-01  4.948e-02  -2.208   0.0391 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.219 on 20 degrees of freedom
## Multiple R-squared:  0.2327, Adjusted R-squared:  0.156
## F-statistic: 3.033 on 2 and 20 DF,  p-value: 0.07071
```

```r
capture.output(
  summary(model),
  file = "results/preliminary_regression.txt"
)
```

The multiple linear regression model examines construction spending and employment together as predictors of poverty rates. The model has an R-squared value of 0.2327, meaning it explains approximately 23.3% of

the variation in poverty rates in our dataset. Employment has a negative coefficient and a p-value of 0.0391, which indicates a statistically significant negative association at the 5% level. Construction spending has a positive coefficient and a p-value of 0.2933, which is not statistically significant.

These findings suggest that employment is more strongly associated with poverty rates than construction spending in our preliminary model. However, the overall model has a p-value of 0.0707, which is above 0.05. With only 23 annual observations, the results should be interpreted carefully. The model also needs further testing before we can determine whether it can reliably predict future poverty rates.

9. Plans for Further Analysis

For the next stage of our project, we plan to examine whether construction spending and employment can reliably predict poverty rates in the United States.

First, we will further evaluate our multiple linear regression model by examining its coefficients, R-squared value, and statistical significance. We will also examine the relationship between our two explanatory variables because construction spending and employment have a relatively strong positive correlation.

Second, we plan to investigate whether changes in construction spending and employment are associated with changes in poverty rates in later years. This may help us understand whether economic changes have delayed relationships with poverty.

Our current results suggest that employment has a stronger relationship with poverty than construction spending.